A PROPER SPACE OF FIBRE DIMENSION AT MOST ONE OVER A COMPLETE NOETHERIAN LOCAL RING IS A PROJECTIVE SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10395-10401 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 10403-10412, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The curve case of the spaces program to recognize schemes over complete local rings; both inputs are chapter internals in this lane's queue area (already published tags, not pool labels).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas formal-algebraic-space-proper-reldim-1 (formal projective approximation of proper spaces of relative dimension <= 1 over complete local rings) and algebraize-morphism (algebraization of compatible system isomorphisms).

Source reference. Original lemma 0A4Z: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10239–10256. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7R: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10339–10353. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a complete Noetherian local ring. Let X be an algebraic space over $\mathrm{Spec}(A)$. If $X \to \mathrm{Spec}(A)$ is proper and $\dim(X_\kappa) \leq 1$, then X is a scheme projective over A.*

Proof. Set $X_n = X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/\mathfrak{m}^n)$. By Lemma 0E7R there exists a projective morphism $Y \to \mathrm{Spec}(A)$ and compatible isomorphisms $Y \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/\mathfrak{m}^n) \cong X \times_{\mathrm{Spec}(A)} \mathrm{Spec}(A/\mathfrak{m}^n)$. By Lemma 0A4Z we see that $X \cong Y$ and the proof is complete. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.